

STUDY GUIDE

The Market Research Analyst Handbook

A complete, concept-first reference — prepared for Saikumar Kaleru.

certifications / study / Finance • 2026

Prepared for Saikumar Kaleru. Read top to bottom; each part builds on the last. Bold terms are the vocabulary interviewers listen for. Indian-market context (CMIE, RBI, SEBI, NSS, industry bodies) is used throughout, matching the rest of this study library.

PART 1 — WHAT MARKET RESEARCH ACTUALLY IS

1.1 The core idea

Market research is the systematic collection, analysis, and interpretation of data about a market — its customers, competitors, and environment — to reduce uncertainty before a business decision. Where **equity research** answers "is this stock worth buying," market research answers "will this product sell, to whom, at what price, and against whom."

Every market research project ultimately answers one of four questions: 1. **Who** is the customer, and what do they need? (segmentation, needs) 2. **How big** is the opportunity? (market sizing) 3. **Who else** is competing for it? (competitive intelligence) 4. **What should we do** about price, product, or positioning? (decision support)

1.2 Market Research vs adjacent disciplines

- **Equity/Investment Research** studies a *company* to price its stock — outward-facing to public markets, ends in a buy/sell call.
- **Market Research** studies a *market or customer base* to guide product/business decisions — ends in a recommendation to a brand, product, or strategy team, not a trade.
- **Competitive Intelligence (CI)** is a specialised slice of market research focused specifically on competitors' moves, capabilities, and likely responses.
- **Business/Data Analytics** answers "what happened" from a company's own internal data; market research goes *outside* the company to answer "what's happening in the market."
- In practice these overlap heavily — a strong market research analyst reads company filings and does light equity-research-style work, and a strong equity analyst does light market-sizing work. The skill sets are transferable, which is the bridge worth using if coming from a finance/markets background.

1.3 Where market researchers sit

- **Agency side:** research/insights consultancies (Nielsen, Kantar, IMRB/Kantar IMRB, Ipsos, GfK, YouGov, Mintel) — client-facing, run studies for many brands.
- **Client/in-house side:** a consumer or B2B company's own insights/strategy team (FMCG, banking, telecom, e-commerce) — owns a category and runs research for internal stakeholders.
- **Consulting:** strategy/management consultancies run market-sizing and competitive-landscape workstreams as part of larger engagements.
- **Financial services angle** (the natural bridge from an equity/derivatives background): buy-side and consulting shops increasingly employ "market intelligence" analysts who combine public-market data with primary research to validate investment theses — this is the most direct crossover role.

1.4 The research process, end to end

1. **Define the business problem** — not the research question yet. ("Should we launch in Tier-2 cities?" is a business problem; "what is unaided brand awareness in Tier-2 cities?" is a research question.)
2. **Translate into research objectives** — specific, answerable questions.
3. **Choose a design** — exploratory, descriptive, or causal (Part 2).
4. **Choose methods** — secondary first (cheap, fast), then primary if gaps remain (Parts 3-4).
5. **Collect data** — fieldwork, surveys, interviews, desk research.

6. **Analyse** — statistical and qualitative analysis (Part 5).

7. **Report and recommend** — translate findings into a decision-ready recommendation (Part 8).

Trap: a research design that answers the *literal* question but not the *business* problem is a common interview failure point — always restate the business decision the research will inform before designing the study.

PART 2 — RESEARCH DESIGN

2.1 The three designs

- **Exploratory research** — used when the problem is vague. Goal: generate hypotheses, not test them. Methods: literature review, expert interviews, focus groups, case analysis. Small samples, qualitative, flexible.
- **Descriptive research** — used to describe a market "as it is": size, share, demographics, usage patterns, satisfaction levels. Methods: surveys, observational studies, secondary-data analysis. Larger samples, mostly quantitative, structured.
- **Causal research** — used to test whether X causes Y (e.g., does a price cut increase volume enough to grow revenue). Methods: experiments, A/B tests, controlled field trials. Requires a control group and random assignment to claim causality — correlation from a plain survey is *not* proof of causation, a point interviewers probe directly.

2.2 Choosing a design — the decision rule

If you don't know what you don't know → **exploratory**. If you need to size or describe something known → **descriptive**. If you need to prove a specific action will change an outcome → **causal**. Most real projects are a funnel: exploratory (a few interviews) → descriptive (a large survey) → occasionally causal (a pilot/A-B test) before a big launch decision.

2.3 Primary vs secondary research

- **Secondary research** uses data that already exists (someone else collected it for another purpose). Cheaper, faster, but not tailored to your exact question and may be stale.
- **Primary research** is data collected specifically for this project (surveys, interviews, experiments). Expensive and slow, but exactly fits the question and is proprietary.
- **Rule of thumb**: always exhaust secondary sources first — most "we need a survey" requests are 60-80% answerable from existing reports, company filings, and government data. This is the single biggest cost-saving move a junior researcher can make, and interviewers listen for it.

2.4 Qualitative vs quantitative

- **Qualitative** — depth over breadth. Answers *why* and *how*. Small samples (6-30), non-statistical, rich narrative data (interview transcripts, discussion notes).
 - **Quantitative** — breadth over depth. Answers *how many*, *how much*, *what %*. Large samples (100s-1000s), statistical, numeric data (survey responses, scanner data).
 - **Mixed-methods** is standard practice: qualitative first to discover the right questions and language, quantitative second to measure how widespread the finding is across the population.
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PART 3 — SECONDARY RESEARCH (DESK RESEARCH)

3.1 Where to look, in order of cost

1. **Company's own data** — internal sales, CRM, support tickets, prior research archives (free, but easy to miss if not asked for).
2. **Government and regulatory sources** (India) — MOSPI/NSS surveys (consumer expenditure, employment), RBI (household finance, banking data), SEBI (mutual fund/market data), Census, Economic Survey, ministry reports (e.g. DPIIT for startups) — free and authoritative, but slow-moving and broad.
3. **Industry bodies and trade associations** — CII, FICCI, NASSCOM (IT), ASSOCHAM, and sector bodies (e.g. AMFI for mutual funds, IBA for banking) — free/cheap, sector-specific, often annual.
4. **Syndicated research firms** — Nielsen (retail audit, FMCG), Kantar, CMIE (Centre for Monitoring Indian Economy — the single most-used paid macro/company database in India), Euromonitor, Statista, Gartner/Forrester (tech), IDC. Paid, but standardised, comparable across companies, and update regularly.
5. **Public company filings** — annual reports, investor presentations, credit-rating rationale (CRISIL/ICRA/CARE), stock-exchange disclosures. Free, primary-sourced by the company itself, most useful for competitive benchmarking of listed players.
6. **News, trade press, and analyst notes** — free, current, but requires triangulation since a single source may be biased or wrong.

3.2 Desk research technique

- **Triangulate:** never trust one source for a market-size or growth number — cross-check at least two independent sources and reconcile the gap explicitly in the report (don't hide the disagreement).
 - **Check the definition, not just the number:** two "market size" figures can differ 5x because one includes adjacent categories (e.g. "EV market" including two-wheelers vs only cars) — always confirm scope before comparing.
 - **Track vintage:** a 2019 number quoted in 2026 without an update flag is a common analyst error and an easy interviewer trap ("how current is this number, and how do you know?").
 - **Build a source log** as you go — every number in the final deck should be traceable to a named, dated source; this is what separates a credible market research deliverable from a guess.
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PART 4 — PRIMARY RESEARCH: QUALITATIVE METHODS

4.1 In-depth interviews (IDIs)

One-on-one, semi-structured conversations (45-60 min), used to explore a topic in depth without group dynamics influencing the answer. Best for sensitive topics (income, health, B2B purchasing decisions) or expert/stakeholder input (e.g. interviewing distributors, doctors, CFOs).

4.2 Focus group discussions (FGDs)

6-10 participants, a trained moderator, 60-90 minutes, used to observe group dynamics, generate ideas through discussion, and test reactions to concepts/ads/packaging live. **Risk:** groupthink and dominant-participant bias — a skilled moderator actively manages this (round-robin questions, private written responses before group discussion).

4.3 Ethnography / observational research

Researchers observe behaviour in its natural setting (a kirana store, a bank branch, a household) rather than asking about it — used because **stated preference and revealed behaviour often diverge** (people say they'd pay for organic food, then buy the cheaper option at the shelf). Slower and more expensive, but the gold standard for behavioural truth.

4.4 Other qualitative tools

- **Projective techniques** — indirect prompts (word association, sentence completion, "if this brand were a person...") used to surface associations people won't state directly.
- **Diary studies** — participants log behaviour/thoughts over days/weeks, useful for infrequent or habitual behaviours a single interview can't capture.
- **Usability/concept testing** — a moderator watches a participant use a product/prototype and think aloud, common in fintech/app research.

4.5 Qualitative analysis

- **Coding:** tagging transcript segments with recurring themes/categories (manually or in NVivo/Atlas.ti).
 - **Thematic analysis:** grouping codes into higher-level themes and reporting prevalence ("6 of 12 respondents mentioned trust as the top barrier") — note this is *illustrative*, not statistically representative, and should never be reported as a percentage of "all customers."
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PART 5 — PRIMARY RESEARCH: QUANTITATIVE METHODS & SURVEY DESIGN

5.1 The survey design process

1. Define exactly what each question will be used to decide (no "nice to know" questions — every question earns its place).
2. Choose question types: **closed-ended** (multiple choice, Likert scale, ranking) for easy quantitative analysis; **open-ended** sparingly, for colour and to catch what closed options missed.
3. Order questions: general → specific, easy → sensitive, behavioural → attitudinal → demographic (demographics last, so people are already invested in finishing).
4. **Pilot test** on 10-20 people before full fielding — catches confusing wording, skip-logic errors, and length problems.

5.2 Common bias traps in question wording

- **Leading questions** — "Don't you agree our app is easy to use?" pushes agreement; ask neutrally: "How easy or difficult is our app to use?"
- **Double-barrelled questions** — "Is the app fast and reliable?" conflates two things; split into two questions.
- **Social desirability bias** — people over-report virtuous behaviour (saving, exercising) and under-report the opposite; mitigate with anonymity and neutral framing.
- **Order/anchoring effects** — the first option in a list gets chosen more often; randomise option order where the survey tool allows it.

5.3 Measurement scales

- **Nominal** — categories with no order (city, gender, brand used).
- **Ordinal** — ordered categories, unequal spacing (income brackets, satisfaction: poor/fair/good/excellent).
- **Interval** — ordered, equal spacing, no true zero (a 1-5 Likert scale is technically interval by convention).
- **Ratio** — ordered, equal spacing, true zero (age, spend in ₹, number of purchases) — allows the widest range of statistics (means, ratios).
- **Likert scale**: the standard 5- or 7-point agreement/satisfaction scale. Interviewers may ask why 5 vs 7 points — 7-point scales give slightly more discrimination but higher respondent fatigue; 5-point is the pragmatic default for most consumer surveys.
- **Net Promoter Score (NPS)**: "How likely are you to recommend X, 0-10?" — % Promoters (9-10) minus % Detractors (0-6), ignoring Passives (7-8). Widely used, widely criticised (a single number hides *why*), always pair with an open-ended follow-up ("what's the main reason for your score?").

5.4 Sampling

- **Population** — everyone the finding should generalise to (e.g. "urban Indian smartphone owners aged 18-35").
- **Sampling frame** — the actual list you can draw from (a customer database, a panel) — rarely a perfect match to the population, and the gap is a source of bias worth naming in any report.

- **Probability sampling** (every unit has a known chance of selection — allows statistically valid generalisation):
- **Simple random sampling** — pure lottery from the full frame.
- **Stratified sampling** — split the population into strata (e.g. by city tier), sample each proportionally — improves precision when subgroups matter.
- **Cluster sampling** — randomly pick clusters (e.g. a few cities), then sample fully within them — cheaper for geographically spread populations, common in India given field-cost constraints.
- **Systematic sampling** — pick every k-th unit from an ordered list.
- **Non-probability sampling** (convenience, quota, snowball, judgment) — faster/cheaper but cannot be generalised with a formal margin of error; acceptable for exploratory work, a red flag if used for a headline market-size claim.

5.5 Sample size — the formula interviewers actually ask for

For a proportion, at 95% confidence:

$$n = (Z^2 \times p \times (1-p)) / e^2$$

Where $Z = 1.96$ (for 95% confidence), p = expected proportion (use 0.5 if unknown — it maximises required sample, the conservative choice), e = desired margin of error (e.g. 0.05 for $\pm 5\%$).

Worked example (India context): A bank wants to estimate the % of urban customers who'd adopt a new UPI feature, $\pm 5\%$ margin of error, 95% confidence, p unknown (use 0.5): $n = (1.96^2 \times 0.5 \times 0.5) / 0.05^2 = (3.8416 \times 0.25) / 0.0025 = 0.9604 / 0.0025 \approx 384$ respondents. This is the famous "384" number that recurs across market research regardless of population size (above a few thousand) — population size barely matters once it's large, which surprises people new to the field and is a good interview talking point.

For a **finite population correction** (when the population itself is small, e.g. surveying a company's 2,000 enterprise clients): $n_{adj} = n / (1 + (n-1)/N)$, where N is the population size.

5.6 Data collection modes

- **CAPI/CATI** (computer-assisted personal/telephonic interviewing) — interviewer-led, good response rates, higher cost, still common for rural/older India respondents.
- **Online panels (CAWI)** — cheapest and fastest, but skews toward internet-literate, younger, urban respondents — a real limitation to disclose, not hide, in any Indian consumer study.
- **Intercept surveys** — stopping people in a physical location (mall, branch, store) — good for in-the-moment behaviour, biased toward that location's footfall profile.

PART 6 — DATA ANALYSIS FOR MARKET RESEARCH

6.1 Descriptive statistics (the first pass on any dataset)

Mean, median, mode, standard deviation, frequency distributions. Always report **median alongside mean** for skewed data (income, spend) — a few high spenders can make the mean misleading.

6.2 Cross-tabulation ("cross-tabs")

Breaking one variable by another (e.g. NPS score by age band) — the single most common output in a market research deck. Always check **base size** (n) in each cell; a finding on a base of 12 respondents is not reportable as a trend, only a footnote.

6.3 Segmentation

Grouping the market into distinct customer types who share needs/behaviour, so different strategies can be aimed at each. - **A-priori segmentation** — segments defined in advance by known variables (age, income, geography) — simple, easy to action, sometimes not the variable that actually drives behaviour. - **Post-hoc (cluster-based) segmentation** — statistical clustering (k-means, hierarchical clustering) on attitudinal/behavioural survey data to *discover* natural groupings — richer, but harder to explain to stakeholders and harder to identify a real customer against ("which cluster is this person?" requires a classifier built on top). - A good segmentation is: **substantial** (big enough to matter), **distinct** (segments actually differ), **stable** (doesn't shift month to month), **actionable** (marketing/product can actually reach and target it).

6.4 Statistical significance testing

- **t-test** — compares means between two groups (e.g. does Segment A spend more than Segment B?).
- **Chi-square test** — tests whether two categorical variables are associated (e.g. is brand preference associated with city tier?).
- **ANOVA** — compares means across 3+ groups at once.
- **p-value**: the probability of seeing a difference this large (or larger) if there were truly no difference in the population. $p < 0.05$ is the conventional "statistically significant" threshold — but always pair with **effect size** (how big is the difference, practically) since a huge sample can make a trivial difference "significant."

6.5 Regression and conjoint analysis

- **Linear/logistic regression** — quantifies how much a set of variables (price, awareness, income) predicts an outcome (purchase, satisfaction) — used for driver analysis ("what actually moves NPS").
- **Conjoint analysis** — shows respondents product profiles that vary systematically on attributes (price, brand, features) and asks them to choose/rank; from choices, the model estimates how much each attribute (and each level, e.g. ₹999 vs ₹1499) contributes to preference. The standard technique for **pricing research** and feature prioritisation — a common interview topic for anyone claiming pricing-research experience.
- **MaxDiff (best-worst scaling)** — respondents repeatedly pick the "most" and "least" important item from small sets — a cleaner alternative to asking people to rate 20 things on a 1-5 scale (where everything ends up rated 4-5).

6.6 Tools of the trade

- **SPSS** — the long-standing standard for cross-tabs, significance testing, and basic multivariate stats in market research; still widely required in job postings.
 - **Qualtrics / SurveyMonkey** — survey design, fielding, and light built-in analysis.
 - **Excel** — pivot tables remain the day-to-day workhorse for cross-tabs and quick analysis.
 - **Power BI / Tableau** — dashboarding for tracking studies (brand trackers, NPS trackers) that update on a cadence.
 - **Python (pandas, statsmodels, scikit-learn)** — increasingly expected for segmentation, regression, and automating repetitive survey-data cleaning — a genuine differentiator for a candidate coming from a Python/SQL background, since most classically-trained market researchers don't code.
 - **R** — still common in academic-leaning or advanced analytics teams for the same statistical work as Python.
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PART 7 — MARKET SIZING & COMPETITIVE INTELLIGENCE

7.1 TAM, SAM, SOM

- **TAM (Total Addressable Market)** — total revenue opportunity if you captured 100% of the relevant market, globally or in the defined geography.
- **SAM (Serviceable Addressable Market)** — the slice of TAM you could realistically serve given your business model/geography/channel constraints.
- **SOM (Serviceable Obtainable Market)** — the slice of SAM you can realistically capture given competition and go-to-market limits, usually over a defined time horizon (e.g. 3 years).

7.2 Two ways to size a market

- **Top-down:** start from a broad, authoritative number (e.g. total Indian retail spend from MOSPI/Census) and apply a chain of narrowing percentages (% online, % in category, % in target city tier) down to the target market. Fast, defensible on data sources, but errors compound multiplicatively across each percentage applied.
- **Bottom-up:** build from unit economics (number of potential customers × price × purchase frequency, or number of outlets × average sales per outlet) — more defensible and specific, but requires more granular data and assumptions to justify.
- **Best practice:** do both and **triangulate** — if top-down and bottom-up land within roughly 20-30% of each other, confidence is high; if they diverge sharply, the discrepancy itself is the most important finding to investigate before presenting either number.

Worked example (India, illustrative): Sizing the market for a premium D2C protein-supplement brand targeting urban gym-goers. - *Top-down:* India population ~1.4bn → urban ~35% (~490m) → adults 18-40 ~30% of urban (~147m) → gym-goers/fitness-conscious ~8% (~11.8m) → willing to pay premium price ~15% (~1.8m potential customers) × assumed ₹6,000/year average spend ≈ **₹1,060 crore SAM**. - *Bottom-up:* ~35,000 organised gyms in India × average 150 active members × 20% buy protein supplements × ₹6,000/year ≈ **₹630 crore**. - The gap (₹1,060cr vs ₹630cr) would be flagged and investigated — likely the top-down "fitness-conscious %" assumption is too generous relative to the bottom-up gym-membership-based count, a classic and defensible finding to walk an interviewer through.

7.3 Competitive intelligence (CI)

- **Competitor mapping:** identify direct competitors (same product, same customer), indirect competitors (different product, same need), and potential entrants.
- **Porter's Five Forces** (also core to equity/strategy work — the crossover point with fundamental research): threat of new entrants, bargaining power of buyers, bargaining power of suppliers, threat of substitutes, competitive rivalry — used to assess how attractive/defensible a market structure is.
- **Win-loss analysis:** systematically interviewing recently won and lost deals/customers to understand *why* — one of the highest-value, most underused CI techniques in B2B contexts.
- **Benchmarking:** comparing pricing, features, service levels, and public financials (for listed competitors) side by side — this is where equity-research-style spreadsheet skills transfer directly into a market research/CI role.

- **Sources for CI in India:** competitor annual reports and investor decks, MCA filings (Ministry of Corporate Affairs — company financials for private Indian companies), app-store reviews and ratings, job postings (signal hiring direction/expansion plans), patent filings, News/PR, LinkedIn headcount trends by function.

7.4 Industry/sector analysis frameworks

- **PESTEL** — Political, Economic, Social, Technological, Environmental, Legal factors shaping an industry's outlook, used for the macro layer of an industry report.
 - **Value chain analysis** — mapping who captures value at each stage (raw material → manufacturing → distribution → retail → customer) to identify where a new entrant or investment could capture margin.
 - **Life-cycle stage** — is the industry in introduction, growth, maturity, or decline? Growth-stage industries reward market-share-gain strategies; mature industries reward cost/efficiency strategies — this framing should show up explicitly in any industry-analyst-style report or interview answer.
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PART 8 — CONSUMER INSIGHTS & REPORTING

8.1 From data to insight

A **finding** is a fact from the data ("62% of respondents cited price as the top purchase barrier"). An **insight** explains *why it matters* and *what to do* ("price sensitivity is highest among first-time buyers, not repeat buyers — a starter SKU at a lower price point could convert this segment without discounting the core range"). Interviewers and hiring managers explicitly screen for candidates who default to insight, not just findings — this is the single most-repeated piece of feedback in market research interview prep material.

8.2 Customer journey mapping

Documenting the stages a customer goes through (awareness → consideration → purchase → onboarding → usage → advocacy/churn), the touchpoints at each stage, and the pain points/emotions at each — used to pinpoint exactly where research or product intervention should focus.

8.3 Personas

Composite, research-based archetypes representing a segment ("Priya, 28, first-jobber, price-sensitive, discovers brands via Instagram") — used to make segmentation memorable and actionable for product/marketing teams. Must be grounded in actual data (segmentation output, real quotes), not invented from assumption — a common quality failure point.

8.4 Reporting and storytelling

- **Structure:** Executive summary (the "so what" in one page) → methodology → key findings (3-5 headline insights, not 40 slides of data) → detailed appendix.
 - **One slide, one message** — a chart should support a single stated takeaway in the slide title, not force the reader to interpret it themselves.
 - **Show the uncertainty:** sample sizes, margins of error, and data vintage belong on every relevant slide, not buried in an appendix — this is what separates rigorous market research from marketing collateral dressed up as research.
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PART 9 — CAREERS, ROLE EXPECTATIONS & INTERVIEW PREP

9.1 Typical role ladder

Research Executive/Analyst (0-2 yrs, fieldwork + basic analysis) → Senior Research Analyst/Associate (2-5 yrs, owns a study end-to-end, client/stakeholder-facing) → Research Manager (5-8 yrs, owns a category/client relationship, manages analysts) → Insights Director/Head of Insights (8+ yrs, sits on the leadership team, sets the research agenda).

"Senior Market Research Analyst" bar (typically 3-5 yrs): can independently scope a study, choose the right method without guidance, run analysis in SPSS/Excel/Python, and present findings directly to a business stakeholder with a clear recommendation — not just execute a brief handed down from a manager.

9.2 What interviewers actually probe for

1. **Methodology judgment** — given a vague business problem, can you design the right study (this handbook's Parts 2-5)?
2. **Numerical fluency** — sample size, significance, basic segmentation logic, market sizing math (Parts 5-7), often via a live case/whiteboard exercise.
3. **"So what" thinking** — can you turn a data table into a decision-ready recommendation (Part 8)?
4. **Domain/business acumen** — do you understand the category (FMCG, banking, fintech, retail) well enough for the insight to be credible, not generic?

9.3 Sample case-style interview question (worked)

"A bank wants to know whether to launch a co-branded credit card with an e-commerce platform. How would you research this?" - Restate the business decision: launch or not, and if yes, target segment and card benefits. - Design: exploratory (a few interviews with existing co-brand-card holders at competitor banks, desk research on comparable co-brand card economics) → descriptive quantitative survey (interest/intent, preferred benefits, willingness to pay annual fee) on a representative sample of the platform's existing customers, stratified by spend tier. - Sizing: bottom-up from the platform's active user base × estimated card-eligible % (income/credit-score proxy) × estimated adoption rate from comparable launches. - Competitive: benchmark 2-3 existing co-brand cards in market (cashback %, fee, partner) via desk research and MITC (Most Important Terms & Conditions) disclosures. - Recommendation: a go/no-go with the target segment, expected uptake, and the single biggest risk flagged explicitly (e.g. cannibalisation of the bank's existing card base). This structure — decision, design, sizing, competitive check, risk-flagged recommendation — is the reusable skeleton for almost any market-research case interview.

9.4 Bridging a finance/derivatives background into market research

The honest gap (see `senior_research_skills.md` in the `job_apply_agent` repo) is direct consumer/survey experience. The honest strength is quantitative rigour most classically-trained market researchers lack: comfort with statistics, sample-size logic, regression, and — critically — **Python for automating survey-data cleaning and analysis**, which is still a differentiator in most agency and in-house insights teams. Frame the F&O/derivatives background as: "trained to read what a large, noisy dataset is actually saying, under time pressure, and to be explicit about uncertainty" — which is the exact same instinct market research analysis requires, applied to a different kind of data.

APPENDIX A — GLOSSARY

TAM/SAM/SOM · **NPS** (Net Promoter Score) · **CATI/CAPI/CAWI** (interview modes) · **Conjoint analysis** · **MaxDiff** · **Cross-tab** · **Base size** · **Significance (p-value)** · **Segmentation (a-priori vs post-hoc)** · **Persona** · **Win-loss analysis** · **CI** (Competitive Intelligence) · **Syndicated research** · **Panel** · **Likert scale** · **Triangulation**

APPENDIX B — KEY FORMULAS

- Sample size (proportion, 95% CI): $n = 1.9604 \times p(1-p) / e^2$
- Finite population correction: $n_{adj} = n / (1 + (n-1)/N)$
- NPS = %Promoters (9-10) – %Detractors (0-6)
- Market size (bottom-up): # potential customers × price × purchase frequency
- Margin of error at 95% CI (given n, p): $e = 1.96 \times \sqrt{p(1-p)/n}$

APPENDIX C — PRACTICE / INTERVIEW Q&A

1. **Q: Why triangulate top-down and bottom-up sizing instead of picking one?** A: Each method has independent failure modes (top-down compounds percentage-assumption errors; bottom-up depends on complete unit-economics data) — agreement between the two is the actual evidence of a credible number, not either method alone.
2. **Q: A survey shows 70% of respondents "would definitely buy" a new product. Should the client trust this for a launch decision?** A: No — stated purchase intent systematically overstates actual purchase behaviour (a well-known market research finding); use intent as a directional signal, and where possible validate with a smaller real-world pilot/test market before a full launch.
3. **Q: When would you use qualitative research instead of a survey, even though surveys are more "statistical"?** A: When the goal is to discover *why* or generate hypotheses (exploratory stage), when the topic is sensitive/hard to pre-code into closed questions, or when sample access is limited (e.g. B2B decision-makers) — depth beats breadth in these cases.
4. **Q: What's wrong with reporting NPS alone as the health metric for a product?** A: It compresses a lot of information into one number and hides *why* scores are what they are; always pair with a verbatim follow-up and driver analysis (Part 6.5) so it's actionable, not just a scoreboard number.
5. **Q: How would you segment the customer base of a digital lending app?** A: Start a-priori (income band, loan purpose, city tier) for quick actionability, then validate/refine with post-hoc clustering on behavioural data (repayment behaviour, app usage frequency, cross-sell uptake) to check whether the a-priori segments actually predict the outcomes that matter (default risk, LTV) — if not, re-cut using the behavioural clusters instead.